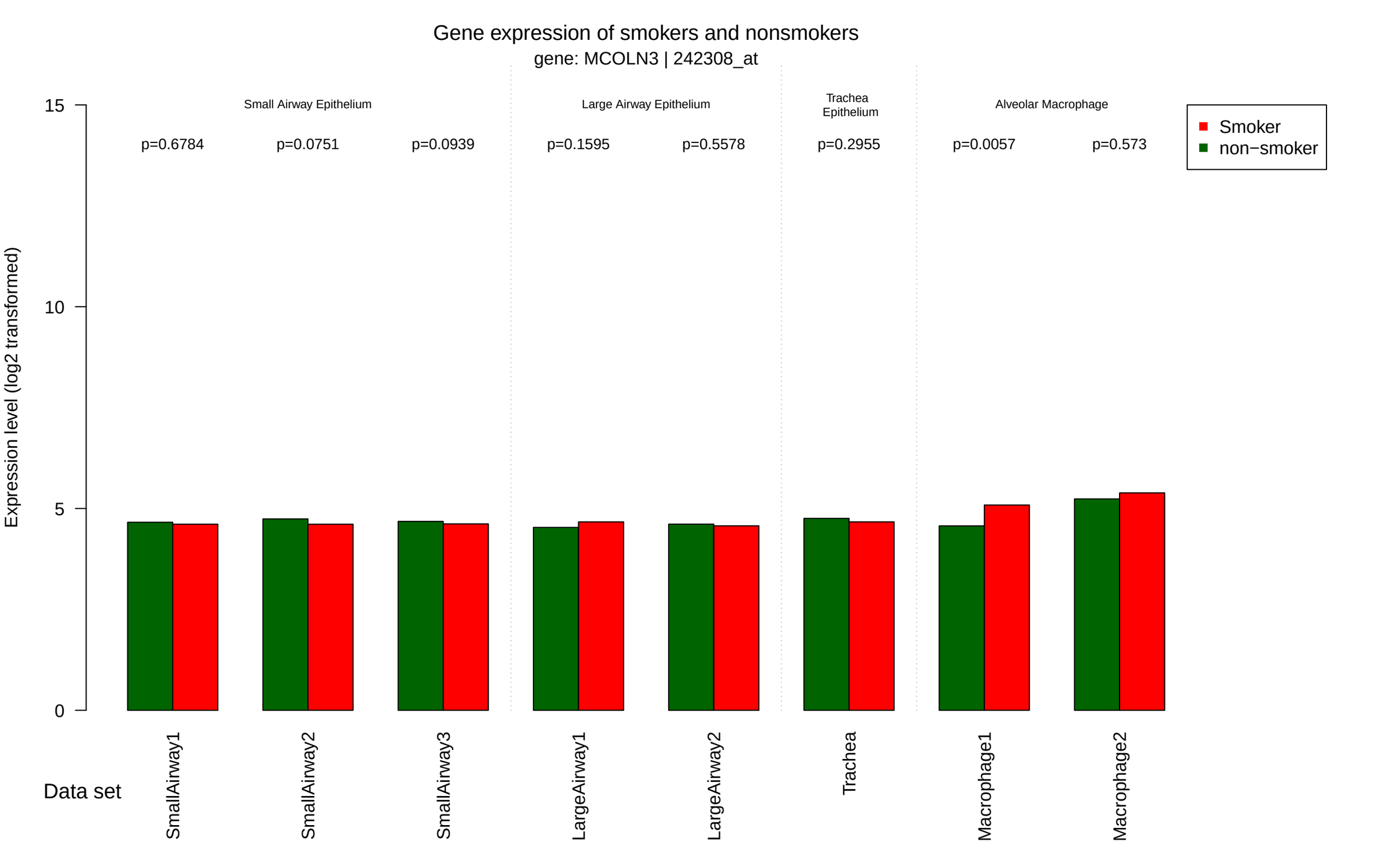
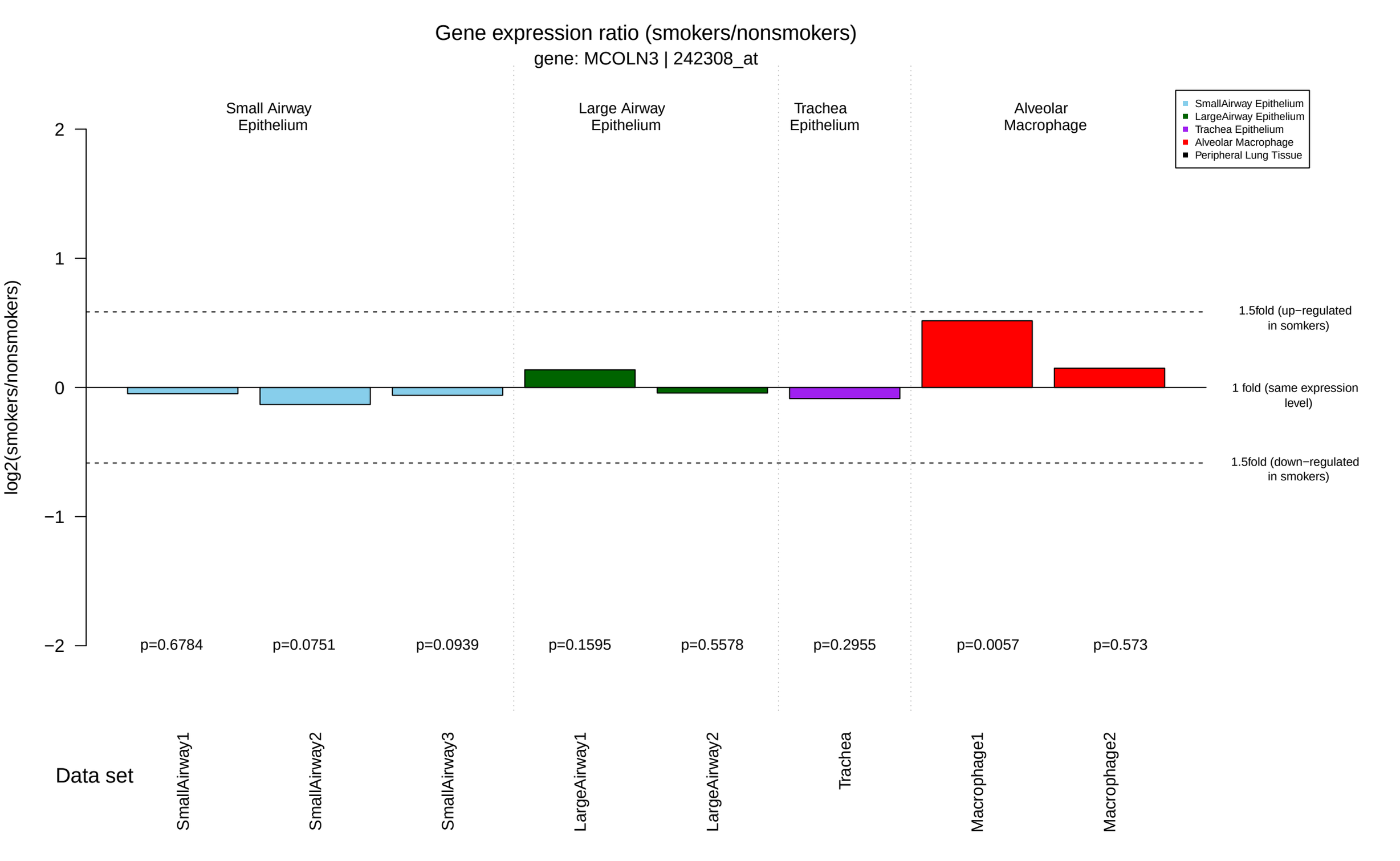
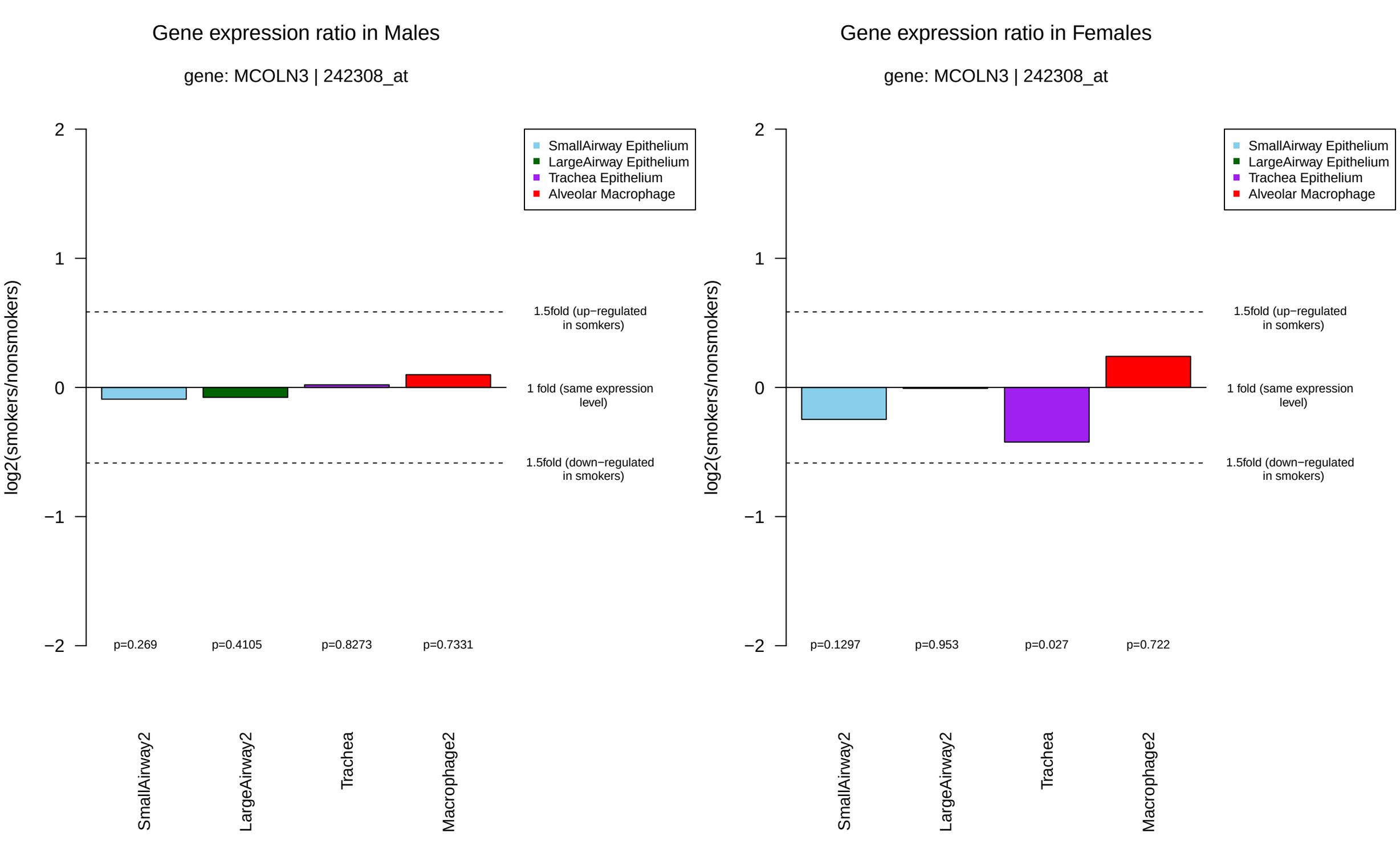
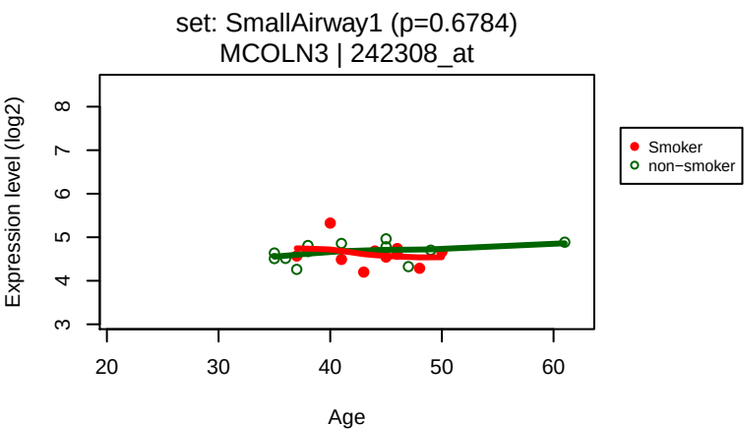
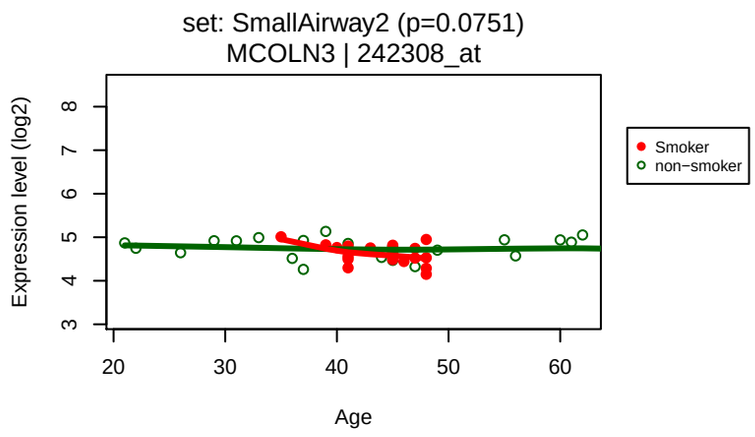
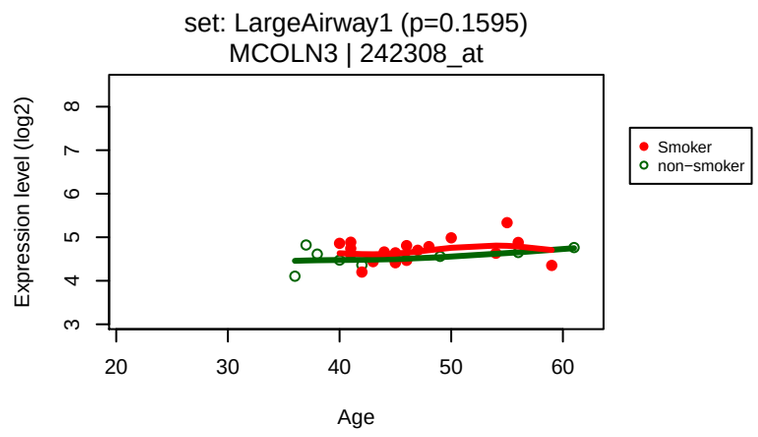

Gene expression of smokers and nonsmokers
gene: MCOLN3 | 242308_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6784
p=0.0751
p=0.0939
p=0.1595
p=0.5578
p=0.2955
p=0.0057
p=0.573
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

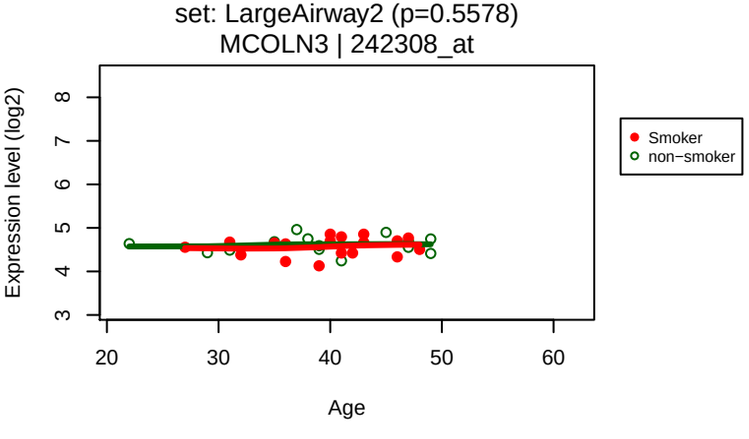

Gene expression ratio (smokers/nonsmokers)
gene: MCOLN3 | 242308_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6784
p=0.0751
p=0.0939
p=0.1595
p=0.5578
p=0.2955
p=0.0057
p=0.573
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

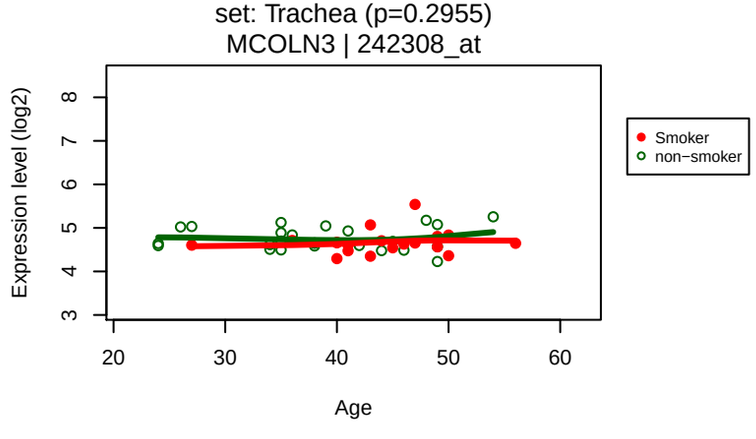

Gene expression ratio in Males
gene: MCOLN3 | 242308_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.269
p=0.4105
p=0.8273
p=0.7331
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MCOLN3 | 242308_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1297
p=0.953
p=0.027
p=0.722
SmallAirway2
LargeAirway2
Trachea
Macrophage2

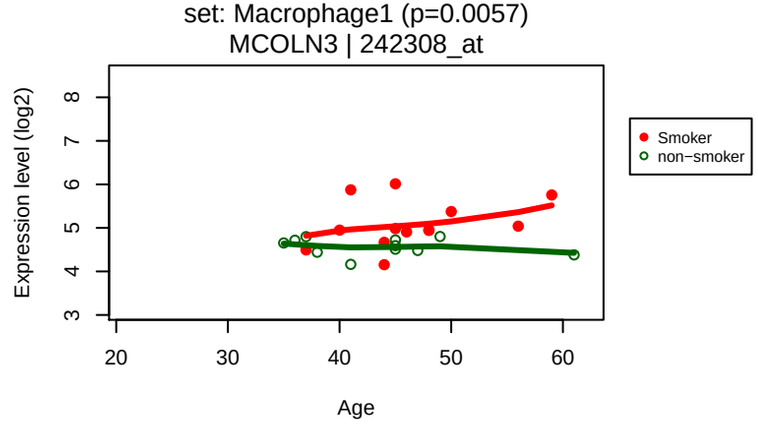

set: SmallAirway1 (p=0.6784)
MCOLN3 | 242308_at
Expression level (log2)
Age
Smoker
non−smoker

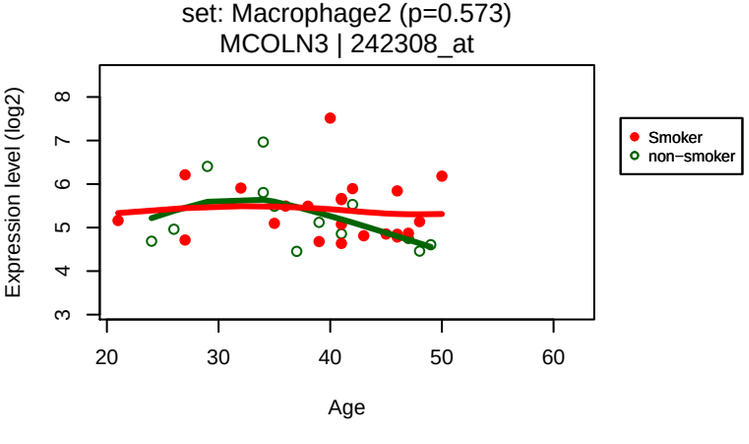

set: SmallAirway2 (p=0.0751)
MCOLN3 | 242308_at
Expression level (log2)
Age
Smoker
non−smoker

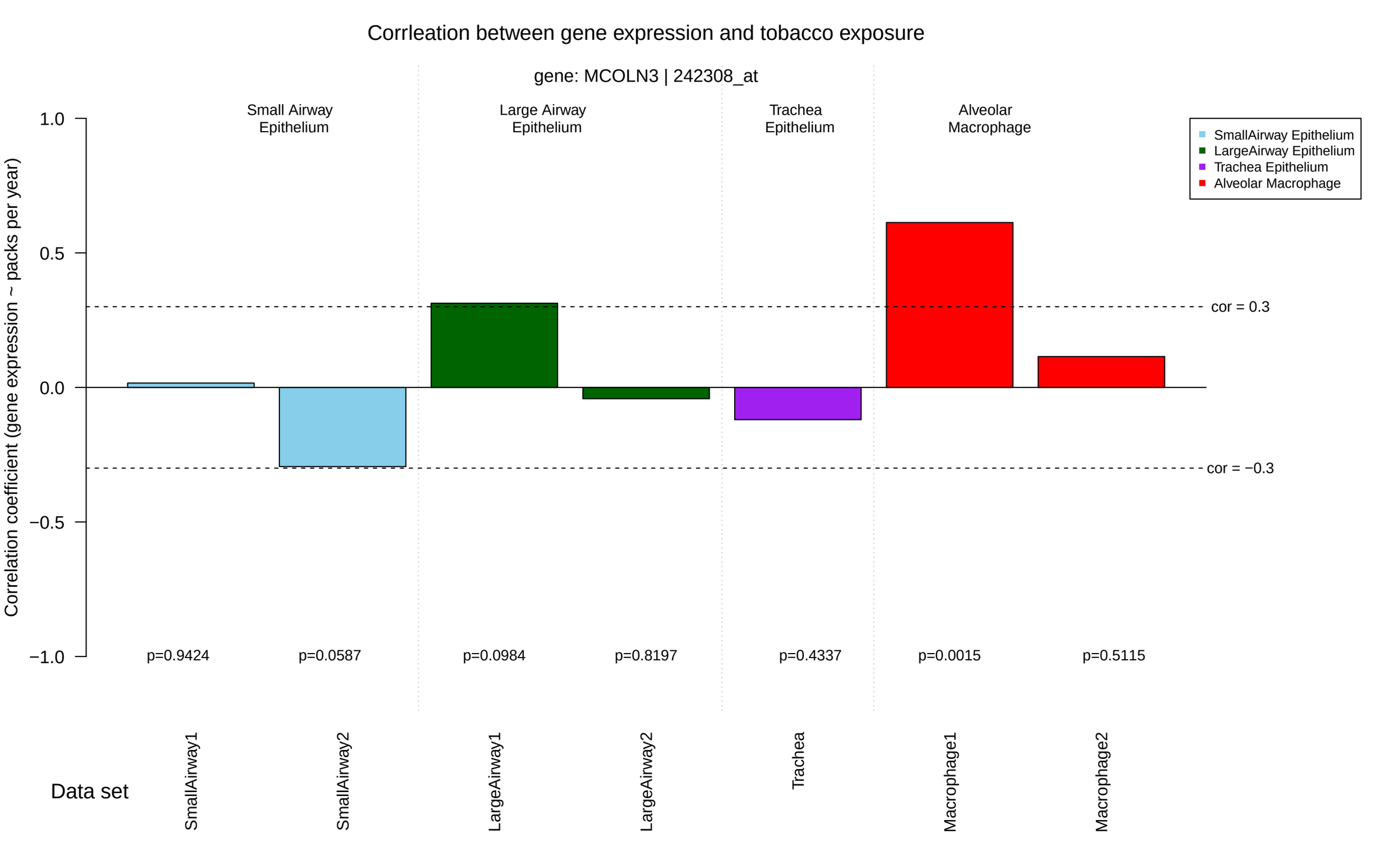

set: LargeAirway1 (p=0.1595)
MCOLN3 | 242308_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5578)
MCOLN3 | 242308_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2955)
MCOLN3 | 242308_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0057)
MCOLN3 | 242308_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.573)
MCOLN3 | 242308_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MCOLN3 | 242308_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9424
p=0.0587
p=0.0984
p=0.8197
p=0.4337
p=0.0015
p=0.5115
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2